Analysis of op T on f.dim vspace V over any field F.

Aim Find a good basis of V so matrix of T is as simple as possible

- Nice to get T-invariant subspace W $\rightsquigarrow$ $T = \begin{pmatrix} T|_W & * \\ 0 & * \end{pmatrix}$
- Nicer to get W and W' T-invariant with $V = W \oplus W'$: $T = \begin{pmatrix} T|_W & 0 \\ 0 & T|_{W'} \end{pmatrix}$ → T breaks up into two separate operators
- Ultimate in this direction get eigenbasis if possible! Mechanics:

T is diagonal $\iff V = \bigoplus_{\lambda \in F} \ker(T - \lambda I)$ $\rightsquigarrow$ λ-espace "V_λ"

λ is an evalue $\iff \ker(\lambda I - T) \neq 0$
$\iff \det(\lambda I - T) = 0$
$\iff \lambda$ is root of char poly of T
namely $P_T(x) = \det(xI - T)$

① $P_T(x)$ factors fully $\iff V = \bigoplus_{\lambda \in F} \ker(T - \lambda I)^{\infty}$ $\rightsquigarrow$ generalized λ-espace "V_λ^{gen}"

② Each (macro summand) gen. eigenspace $V_\lambda^{gen} = \bigoplus$ T-stable subspaces W_i such that $T|_{W_i}$ has a Jordan basis i.e. matrix of $T|_{W_i}$ wrt this magic basis has form $\begin{bmatrix} \lambda & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{bmatrix}$.

Proof of ②: easily reduces to showing for any nilpotent T a Jordan basis! Proof for nilpotent T. later (see the napkin)

③ Uniqueness: $\dim V_{\lambda_i}^{gen} = m_i \iff P_T(x) = \prod_i (x-\lambda_i)^{m_i}$

and for each i, the sizes of Jordan blocks within $T|_{V_{\lambda_i}^{gen}}$ are uniquely decided.

(Two matrices in J form are conjugate/similar $\iff$ The multiset of sizes of Jordan blocks for each eigenvalue is the same)

Tools · $P_T(x) = (x-\mu_1)\cdots(x-\mu_n) \iff \exists$ a basis of V wrt which matrix of T is upper $\triangle$ with diag entries $\mu_1, \cdots, \mu_n$

· T = operator on V of dim n over field F leads to chains of T-stable subspaces

$$0 \subset \ker T \subset \ker T^2 \subset \quad \subset \ker T^n = \ker T^\infty$$
$$V \supset \operatorname{im} T \supset \operatorname{im} T^2 \supset \cdots \supset \operatorname{im} T^n = \operatorname{im} T^\infty$$

} that stabilize as soon as a term repeats, so certainly by term n

Fitting decomposition: $V = \ker T^n \oplus \operatorname{im} T^n$

↑ T acts nilpotently

↑ T is an automorphism

· **Recall** T nilpotent on V of dim n
$\iff T^n = 0 \iff P_T(x) = x^n$
$\iff T$ has upper $\triangle$ matrix with 0 on diag

Note: For any scalar $\lambda \in F$ a subspace of V is T-stable $\iff$ it is $(T-\lambda I)$-stable

Pf of ① Consider operator $T-\lambda_1 I = S$ on V Fitting $\Rightarrow V = \ker S^\infty \oplus \operatorname{im} S^\infty$
$= U \oplus W$

Let $P_T(x) = \prod (x-\lambda_i)^{m_i}$

$S = T-\lambda_1 I$ has { a strictly upper $\triangle$ matrix on $\ker S^\infty$; an invertible matrix on $\operatorname{im} S^\infty$

$\therefore T|_U$ has a matrix of form $\begin{bmatrix} \lambda_1 & * \\ 0 & \lambda_1 \end{bmatrix}$

$P_T(x) = P_{T|_U}(x) \; P_{T|_W}(x)$

$= (x-\lambda_1)^{\dim U} \{$ (poly that does NOT have λ_1 as a root)

$\therefore$ By comparison with given factorization of $P_T(x)$ we have $\dim U = m_1$

Note that $P_{T|_W}(x) = \prod_{i>1} (x-\lambda_i)^{m_i}$. Induction finishes the proof

Objection?! Why $\ker (T-\lambda_2 I)^\infty = \ker (T|_W - \lambda_2 I_W)^\infty$ } ok b/c by this each space has $\dim m_2$